GlucoTwin: Multimodal Digital Twin System Architecture

2-Hour Continuous Glucose Spike Prediction & In-Silico Intervention Simulation | Team Aethan, Amity University Mumbai

1. DATA INGESTION

Dual Stream Telemetry

STATIC EHR STREAM (Synthea)
- Demographics (Age, BMI, City)
- Biomarkers (HbA1c, Fasting BG)
- Meds, Comorbidities (46 feats)

DYNAMIC WEARABLE/CGM
- 5-min Interstitial Glucose
- Heart Rate & SDNN HRV
- Pedometer / Activity Mode
- Sleep Stages & Duration
- 1.73M Readings / 200 Pts

2. FEATURE PIPELINE

FeatureStore & Leak-Free Splits

Static Preprocessing (46 feats)
ICMR Asian Indian BMI, TG/HDL, CKD

Temporal Kinetics (54 feats)
- 2h Rolling Mean, Std, Accel, CV%
- Diurnal Circadian & Meal Clocks

Patient-Stratified Splitter
Train (140) | Val (20) | Test (40)
Strict zero-leakage sequence grid
X_temp: (24x54), X_stat: (46)

3. HYBRID AI ENSEMBLE

Calibrated Forecasting Engine

BRANCH A: PyTorch Attention-LSTM (0.6)
- 2-Layer BiLSTM (128 units) + Self-Attn
- Temporal saliency weighting across 2h window

BRANCH B: Tabular XGBoost (0.4)
- 478 Rolling stats + static features
- Asymmetric cost-sensitive scale_pos_weight

GLUCOTWIN ENSEMBLE FUSION
ROC-AUC: 0.993 | PR-AUC: 0.997 | Acc: 95.3%
2-Hour Forward Spike Probability (>180 mg/dL)

4. DIGITAL TWIN ENGINE

In-Silico Virtual Patient

Physiological State Tracker
Live Glycemic Velocity & Basal Targets

Clinical Risk Calculator
ADA / RSSDI Tiering (Low->Critical)

In-Silico What-If Simulator
Counterfactual Trials:
- GLUT4 Brisk Walk (-48 mg/dL)
- Med Titration (-38 mg/dL)

FASTAPI HIGH-PERFORMANCE BACKEND

RESTful Clinical Microservice (0.0.0.0:8000)

- GET /api/patients/:id/dashboard - Consolidated Clinical Payload in <20ms
- GET /api/patients/:id/predict - 2-Hour Trajectory, Spikes, Saliency Attention
- POST /api/patients/:id/simulate - In-Silico Intervention Counterfactuals
- GET /api/patients/:id/alerts - Acute Hypoglycemia & Critical Hyperglycemia Alerts

DPDP Act 2023 & HIPAA Compliant Sandbox | 100% Synthetic Anonymized Telemetry

NEXT.JS 14 CLINICAL TWIN DASHBOARD

Real-Time Physician Telehealth Portal

- PatientOverview - Demographics, Indian Clinical Phenotype, HbA1c, Meds
- GlucoseChart - Recharts 24h Trend + 2h Dashed Forecast with Euglycemic Bands
- RiskGauge & Timeline - Real-time Spike Probability, Time-to-Spike & Feature Drivers
- TwinVisualization - Multi-Organ Strain Meters & Interactive 'What-If' Modal

Dark Medical Aesthetic | Responsive CSS Grid | 30s Real-time Live Polling